

Ethical Rulebook for Autonomous Vehicle Decision-Making

Overview

This project applies two ethical frameworks to guide how an autonomous vehicle (AV) should act in unavoidable crash scenarios. Each framework provides a distinct method for determining the most appropriate action.

Ethical Frameworks Used

- **Utilitarian Ethics**
- **Kantian Ethics**

Available Actions

In every scenario, the AV must choose exactly one of the following actions:

- **Maintain Course:** Continue on the current path without deviation. This may include braking or accelerating.
- **Swerve Left**
- **Swerve Right**

General Assumptions

- Every possible action results in at least one fatality.
 - The system must always select one action. If a tie is present, choose in the order of precedence: maintain course, swerve left, swerve right. (if a tie is present between maintain course and another choice, choose maintain course. If a tie is present between swerve left and swerve right, choose swerve left.)
 - Harm to passengers is considered an intentional outcome of the AV's decision.
 - Maintaining course is interpreted as *not actively intervening*.
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1. Utilitarian Rulebook

Core Principle

Utilitarian ethics requires choosing the action that minimizes total harm. The primary objective is to reduce the total number of fatalities. Every person, regardless of identity, should be treated as equal.

Decision Process

For each possible action:

1. Estimate the number of individuals who would be killed.
2. Calculate the total fatalities for that action.
3. Select the action with the lowest total number of deaths.

Key Insight

Utilitarianism is centered on minimizing overall harm. Therefore, the optimal decision is always the one that results in the smallest number of fatalities.

2. Kantian Rulebook

Core Principle

Kantian ethics emphasizes moral rules and respect for individuals rather than outcomes. The guiding rule is:

Never treat a person merely as a means to an end.

This means the AV must not intentionally sacrifice one individual to save others.

Decision Process

For each possible action, the system evaluates:

1. Does the action intentionally cause a person's death?
2. Does it use a person's death as a means to prevent other deaths?

If the answer to either question is **yes**, the action is considered morally unacceptable.

Key Insight

Kantian ethics prioritizes avoiding intentional harm to individuals, even if doing so would reduce the total number of fatalities.

3. Virtue Ethics Rulebook

Core Principle

Virtue ethics focuses on what a morally good and practically wise agent would do in a given situation. The goal is not only to reduce harm, but to act with **good judgment**, balancing compassion, justice, and responsibility.

Moral Value Ordering

When evaluating harm, individuals are not treated as strictly equal. The relative moral priority of pedestrians is:

Pregnant > Child > Disabled > Elderly > Adult

Decision Process

For each possible action:

Step 1: Evaluate Harm Severity

- Assign each potential fatality a weight based on the moral value ordering:
 - Pregnant > Child > Disabled > Elderly > Adult
- Compute a **total harm score** for each action using these weights

Step 2: Apply Responsibility Consideration

- Maintaining course is considered **passive**
- Swerving is considered **active intervention**, which introduces new harm

Apply a **penalty to actions that involve redirection of harm** (swerve left or right), unless justified by a significant reduction in harm.

Step 3: Apply Practical Judgment (Prudence)

- If one action results in **substantially lower total harm**, select that action
- If differences in harm are small or comparable:
 - Prefer **maintaining course** to avoid unnecessary intervention
- If harm is equal across actions:
 - Follow the standard tie-breaking rule